# Fundamental Trigonometric Identities

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.02

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

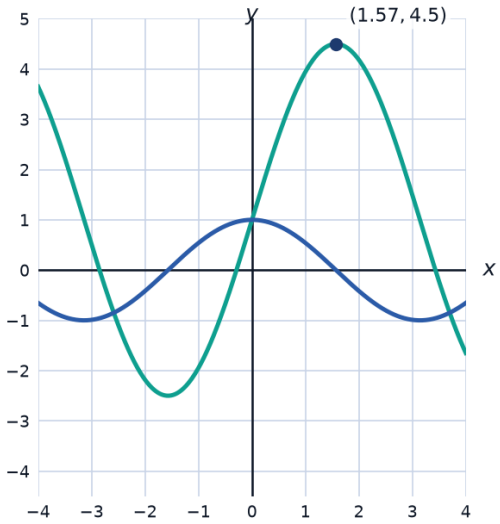
**1** [Foundational] • Target: Using the Pythagorean identities

Given that $\sin\theta = \frac{3}{5}$ and $\theta$ lies in Quadrant II, find:

**a** $\cos\theta$ **b** $\tan\theta$ **c** $\sec\theta$ **d** $\csc\theta$

**2** [Foundational] • Target: Using the Pythagorean identities

Given that $\tan\theta = 2$ and $\theta$ lies in Quadrant III, find $\sin\theta$ and $\cos\theta$ in exact simplified form.




**3** [Foundational] • Target: Simplifying and verifying

Simplify each expression to a single trigonometric function:

**a** $\frac{1-\cos^2 x}{\sin x}$

**b** $\sec x \cot x$

**c** $\tan x \csc x$

**4** [Foundational] • Target: Simplifying and verifying

Verify the identity $(1+\tan^2 x)\cos^2 x = 1$.

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**5** [Foundational] • Target: Quotient and reciprocal identities

Rewrite each expression using only $\sin x$ and $\cos x$, then simplify:

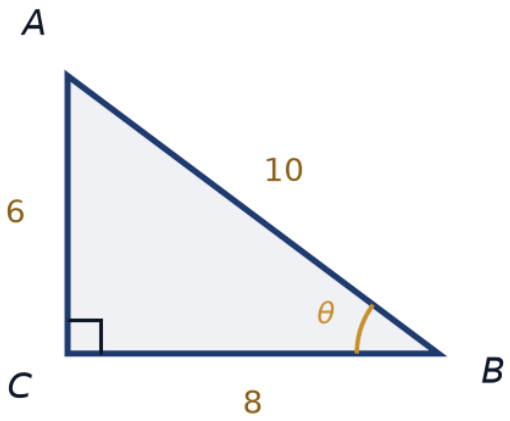
**a** $\cot x \sin x$

**b** $\frac{\sec x}{\csc x}$

**c** $1 - (\sin^2 x \csc^2 x)\sin^2 x$

**6** [Foundational] • Target: Quotient and reciprocal identities

Simplify $\frac{\cos x}{1 - \sin x} - \tan x$ to a single term.




**7** [Foundational] • Target: More identity verifications

Verify the identity $\sec^2 x - \tan^2 x = 1$ using the Pythagorean identity for tangent and secant.

## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** [Intermediate] • Target: Identities with sums and products

Simplify $(\sec x - \tan x)(\sec x + \tan x)$ to a constant.

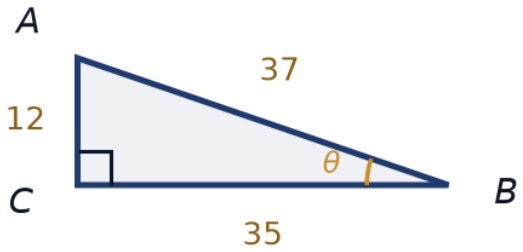
**9** [Intermediate] • Target: Identities involving all three Pythagorean forms

State all three forms of the Pythagorean identity (for sine/cosine, tangent/secant, and cotangent/cosecant).

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**10** [Intermediate] • Target: Identities involving all three Pythagorean forms

Given $\cot\theta = \frac{5}{12}$ and $\theta$ in Quadrant I, find $\csc\theta$ and $\sin\theta$ using the cotangent-cosecant identity.




**11** [Intermediate] • Target: Simplifying more complex expressions

Simplify $\frac{\sin^3 x + \cos^3 x}{\sin x + \cos x}$ using the sum-of-cubes factoring pattern.

**12** [Intermediate] • Target: Numeric identity verification (checking a specific angle)

Verify the identity $1 + \tan^2 x = \sec^2 x$ numerically at $x = \frac{\pi}{6}$ by computing both sides.

**13** [Intermediate] • Target: Identities with squared expressions

Given that $\sec\theta = -\frac{13}{5}$ and $\theta$ lies in Quadrant II, use the Pythagorean identity $1 + \tan^2\theta = \sec^2\theta$ to find $\tan\theta$ exactly, and then find $\sin\theta$ using the quotient identity.

**14** [Intermediate] • Target: Identities with squared expressions

Verify the identity $\frac{\csc x - 1}{\cot x} = \frac{\cot x}{\csc x + 1}$ by cross-multiplying and using the Pythagorean identity.

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

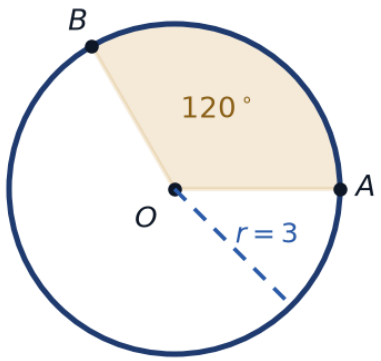
**15** [Challenging] • Target: Identities with sums and products • Cross-Lesson: 8.01 Rational Functions and Asymptotes

Find all solutions to the trigonometric equation $2\sin^2 x - 3\sin x + 1 = 0$ on the domain $x \in [0, 2\pi)$.

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**16** [Challenging] • Target: Identities involving all three Pythagorean forms • Cross-Lesson: 8.02 Graphing Rational Functions

[Unit Circle & Trigonometric Identities · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input $t$ when $g(t) = 50$.




**17** [Challenging] • Target: Simplifying more complex expressions • Cross-Lesson: 8.03 Introduction to Limits

Rotate the Cartesian axes to eliminate the cross-product term in $5x^2 - 4xy + 8y^2 = 36$. Find the angle of rotation $\theta$, standard form in the rotated system $(x', y')$, and the lengths of the semi-major and semi-minor axes.

**18** [Challenging] • Target: Verifying identities with a common denominator strategy • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Prove by mathematical induction that for all integers $n \geq 1$: $1 + 3 + 5 + \ldots + (2n - 1) = n^2$.

**19** [Challenging] • Target: Numeric identity verification (checking a specific angle) • Cross-Lesson: 9.02 Geometric Sequences and Series

Solve the logarithmic equation $\log_2(x + 3) + \log_2(x - 3) = 4$ for all real solutions $x$. State any extraneous roots.

**20** [Challenging] • Target: Identities with squared expressions • Cross-Lesson: 9.03 The Binomial Theorem

An infinite geometric series has common ratio $r = -0.4$ and converges to an infinite sum of $S_\infty = 25$. Find the first term $a_1$, and compute the exact value of the partial sum $S_4$.

## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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**21** [Foundational] • Target: Using the Pythagorean identities

Divide $f(x) = 2x^3 + 5x^2 + 5x + 2$ by $(x + 1)$ using synthetic division. Find the quotient.

**a.** $2x^2 + 3x + 2$

**b.** $2x^2 + 3x - 2$

**c.** $2x^2 - 3x + 2$

**d.** $3x^2 + 3x + 2$

**22** [Foundational] • Target: Simplifying and verifying

Consider the complex number $z = r(\cos\theta + i\sin\theta)$. If $z^3 = -8$ and $r > 0$, and the principal argument of $z$ is $\frac{\pi}{3}$, find the value of $r$.

**a.** 2

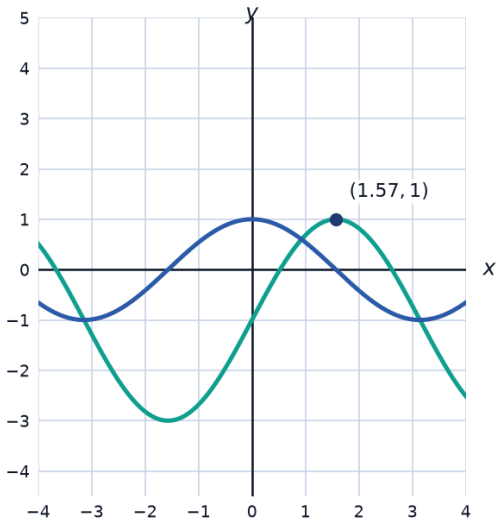
**b.** 4

**c.** 8

**d.** 16

**23** [Foundational] • Target: Quotient and reciprocal identities

Consider the exponential growth model described by the function $P(t) = P_0e^{kt}$, where $P(t)$ is the population at time $t$, $P_0$ is the initial population, and $k$ is the growth constant. If the initial population $P_0$ is 500 and the population after 4 years is 2000, determine the value of the growth constant $k$ to three decimal places.




**a.** 0.347

**b.** 0.416

**c.** 0.301

**d.** 0.477

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**24** [Foundational] • Target: More identity verifications

A complex number $z$ is represented in polar form as $z = r(\cos(\theta) + i\sin(\theta))$. If a complex number has a modulus $r = 5$ and an argument $\theta = \frac{\pi}{3}$, what is its rectangular form $a + bi$?

**a.** $\frac{5}{2} + i\frac{5\sqrt{3}}{2}$

**b.** $5 + i5$

**c.** $\frac{5\sqrt{3}}{2} + i\frac{5}{2}$

**d.** $5i$

**25** [Foundational] • Target: Identities with sums and products

Find all real roots of $f(x) = x^3 - 4x^2 - 36x + 144$.

**a.** x=-6

**b.** x=-6,4

**c.** x=-5,4,6

**d.** x=-6,4,6

**26** [Foundational] • Target: Identities involving all three Pythagorean forms

Find the common difference of the arithmetic sequence: -1, -2, -3, -4, ...

**a.** 0

**b.** -2

**c.** 1

**d.** -1

**27** [Foundational] • Target: Simplifying more complex expressions

Find the 4th term of the geometric sequence with $a_1 = 2$ and common ratio $r = 2$.

**a.** 18

**b.** 17

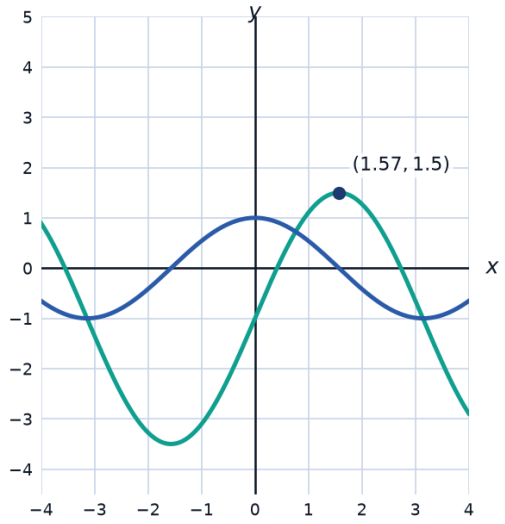
**c.** 32

**d.** 16

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**28** [Foundational] • Target: Verifying identities with a common denominator strategy

Let $(a_n)_{n \geq 1}$ be an arithmetic sequence of real numbers with first term $a_1 > 0$ and non-zero common difference $d$. The terms $a_1$, $a_3$, and $a_4$ form the first three terms of a convergent geometric sequence $(g_k)_{k \geq 1}$, such that $g_1 = a_1$, $g_2 = a_3$, and $g_3 = a_4$. Let $S_\infty = \sum_{k=1}^{\infty} g_k$ denote the sum to infinity of the geometric sequence, and let $S_n = \sum_{j=1}^{n} a_j$ denote the $n$-th partial sum of the arithmetic sequence. If $M = \max_{n \geq 1} S_n$ is the maximum value attained by the partial sums of the arithmetic sequence, what is the value of $\frac{M}{S_\infty}$?




**a.** $\frac{5}{4}$

**b.** $\frac{81}{64}$

**c.** $\frac{9}{8}$

**d.** $\frac{5}{8}$

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: Numeric identity verification (checking a specific angle)

Using a double-angle identity, evaluate $\cos^2\theta - \sin^2\theta$ when $\theta = 15^\circ$.

**a.** -1

**b.** $-\frac{\sqrt{3}}{2}$

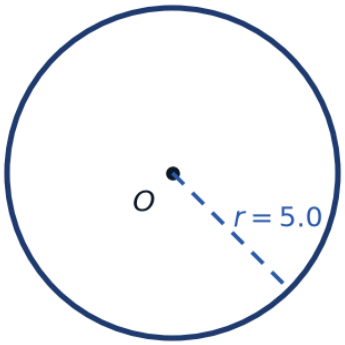
**c.** $\frac{\sqrt{2}}{2}$

**d.** $\frac{\sqrt{3}}{2}$

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**30** [Intermediate] • Target: Identities with squared expressions

Find the center and radius of $(x+6)^2+(y-1)^2=25$.




**a.** Center $(6, -1)$, radius 5

**b.** Center $(-6, 1)$, radius 5

**c.** Center $(1, -6)$, radius 5

**d.** Center $(-6, 1)$, radius 25

**31** [Intermediate] • Target: Using the Pythagorean identities

Using complex numbers, find $\cos^4\theta$ as a sum of cosines.

**a.** $\frac{3+4\cos 2\theta+\cos 4\theta}{8}$

**b.** $\frac{1+\cos 4\theta}{2}$

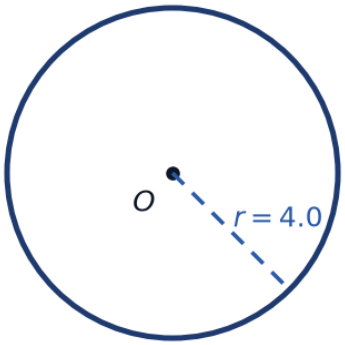
**c.** $3/8$

**d.** $\cos 4\theta/8$

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**32** **[Intermediate]** • **Target: Simplifying and verifying**

What is the center of the circle $(x-3)^2+(y+2)^2=16$?




**a.** (-3, 2)

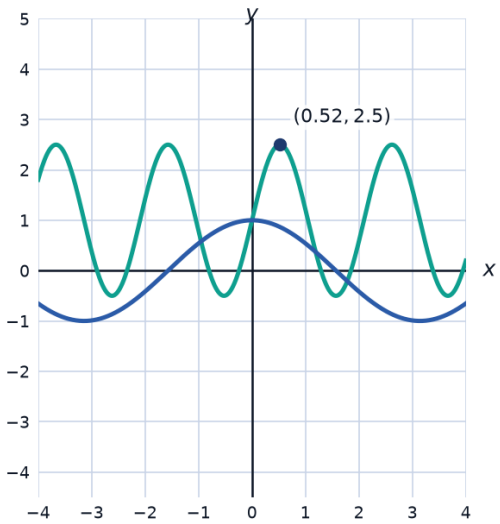
**b.** (3, -2)

**c.** (3, 2)

**d.** (-2, 3)

**33** **[Intermediate]** • **Target: Quotient and reciprocal identities**

For $z=re^{i\theta}$, express $\mathrm{Re}(z^3)$ using triple angle formula.




**a.** $r^3$

**b.** $3r^3\cos\theta$

**c.** $r^3\cos 3\theta$

**d.** $r\cos 3\theta$

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**34** [Intermediate] • Target: More identity verifications

Evaluate $\sum_{k=1}^{n} \cos k\theta$ using complex exponentials.

**a.** $0$

**b.** $n\cos\theta$

**c.** $\frac{\sin(n\theta/2)}{\sin(\theta/2)}\cos\frac{(n+1)\theta}{2}$

**d.** $\cos(n\theta)$

**35** [Intermediate] • Target: Identities with sums and products

State $\cos 2\theta$ in three equivalent forms.

**a.** $\cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$

**b.** $\cos\theta + \sin\theta$

**c.** $2\sin\theta\cos\theta$

**d.** $\cos^2\theta$

**36** [Intermediate] • Target: Identities involving all three Pythagorean forms

A sequence is defined by the recurrence relation $a_{n+1} = a_n^2 - 2$ for $n \geq 1$. If $a_1 = 2$, find the value of $a_{2023}$.

**a.** 2

**b.** 0

**c.** -2

**d.** 1

**37** [Intermediate] • Target: Simplifying more complex expressions

A population of bacteria, initially $P_0$ cells, grows according to the logistic model $P(t) = \frac{L}{1 + Ae^{-kt}}$, where $t$ is in hours, $L$ is the carrying capacity, and $A$ and $k$ are positive constants. At $t = 0$, the population is $P(0) = \frac{L}{3}$. After 2 hours, the population has doubled to $P(2) = \frac{2L}{3}$. Find the exact value of $e^{2k}$ in terms of $L$ and $P_0$.

**a.** $\frac{9}{4}$

**b.** $\frac{4}{9}$

**c.** $3$

**d.** $\frac{3}{2}$

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## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: Verifying identities with a common denominator strategy • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Consider the exponential function $f(x) = 2^x$. If $f(a) = 8$, what is the value of $a$?

**a.** 1

**b.** 2

**c.** 3

**d.** 4

**39** [Challenging] • Target: Numeric identity verification (checking a specific angle) • Cross-Lesson: 9.02 Geometric Sequences and Series

Consider the exponential function $f(x) = 2^x$. What is the value of $f(3)$?

**a.** 6

**b.** 8

**c.** 9

**d.** 5

**40** [Challenging] • Target: Identities with squared expressions • Cross-Lesson: 9.03 The Binomial Theorem

Find the vector projection of $\mathbf{u} = \langle 3, 4 \rangle$ onto $\mathbf{v} = \langle 2, 0 \rangle$.

**a.** $\langle 0, 3 \rangle$

**b.** $\langle 4, 0 \rangle$

**c.** $\langle 3, 0 \rangle$

**d.** $\langle 3, 4 \rangle$

**41** [Challenging] • Target: Using the Pythagorean identities • Cross-Lesson: 1.01 Functions and Function Notation

For $f(x) = \frac{x^2 - 25}{x - 5}$ when $xeq5$, and $f(5) = k$, find $k$ so that $f$ is continuous at $x = 5$.

**a.** 10

**b.** 5

**c.** 9

**d.** 11

**42** [Challenging] • Target: Simplifying and verifying • Cross-Lesson: 1.02 Domain and Range

The arithmetic sequence with $a_1 = 3$ and $d = 4$ has $a_n = 99$. Find $n$.

**a.** 25

**b.** 99

**c.** 26

**d.** 24

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**43** [Challenging] • Target: Quotient and reciprocal identities • Cross-Lesson: 1.03 Average Rate of Change

Solve for the larger value of $x$: $2^{2x} - 10 \cdot 2^{x} + 16 = 0$.

**a.** 5

**b.** 3

**c.** 1

**d.** 4

**44** [Challenging] • Target: More identity verifications • Cross-Lesson: 2.01 Right-Triangle Trigonometry

For the parabola $y^2 = -20x$, find the vertex, focus, and directrix.

**a.** Vertex $(0, 0)$, focus $(-5, 0)$, directrix $x = 5$

**b.** Vertex $(0, 0)$, focus $(-10, 0)$, directrix $x = 10$

**c.** Vertex $(0, 0)$, focus $(0, -5)$, directrix $y = 5$

**d.** Vertex $(-5, 0)$, focus $(0, 0)$, directrix $x = -5$

**45** [Challenging] • Target: Identities with sums and products • Cross-Lesson: 2.02 Radian Measure and Arc Length

Find the $10$th term of the arithmetic sequence with $a_1 = 3$ and $d = 4$.

**a.** 36

**b.** 35

**c.** 39

**d.** 43

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